Chapter 11 — Request Deduplication & Single-Flight

> **Part II — Networking Patterns** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 1 (Networking Core — `safeApiCall`, `Outcome`), Ch 5 (Hilt singletons), Ch 6 (concurrency context). The **single-flight** primitive defined here is reused by Ch 12 (cache stampede), Ch 19 (token refresh), Ch 13 (paging), Ch 59 (media manifest).
>
> **Reference surface in StreamCart:** collapsing duplicate profile/catalog/config fetches issued by multiple screens at once into one in-flight request.

11.1 Problem Definition and Real-World Context

The same request is frequently triggered multiple times, concurrently, for the same data. Without coordination, each trigger becomes a separate network call:

- **Multiple screens, same data.** The home screen, the profile chip, and a background prefetch all request `GET /me` within the same second → three identical calls, three parses, triple backend load.
- **Rapid user actions.** Pull-to-refresh tapped twice, or a button double-tapped, fires two identical mutations/fetches.
- **Cache stampede (thundering herd).** A cached value expires and twenty collectors miss simultaneously; all twenty hit the network for the same key.
- **Token refresh storm.** Fifty queued requests all get 401 at once; naive code triggers fifty refreshes (Ch 19) — the canonical single-flight problem.

Single-flight (a.k.a. request coalescing / de-duplication) ensures that for a given key, **at most one request is in flight**, and all concurrent callers share its single result. It's a small, reusable concurrency primitive with outsized impact on backend load, battery, and correctness.

Three practical use cases

1. **E-commerce config/profile (StreamCart).** Many components need `/me` or `/config` on launch; single-flight collapses them into one call whose result fans out to all.
2. **OTT playback (Watch).** Several UI elements request the same playback manifest/license when a title opens; coalesce to one (Ch 59).
3. **Fintech refresh (Wallet).** A balance refresh triggered by screen-resume and a pull-to-refresh at the same time should hit the backend once.

Why this is architectural

It defines a **coordination primitive** (keyed, concurrency-safe, result-sharing) that multiple features and other infrastructure (caching, auth) build on. Where it lives, how keys are defined, and its cancellation semantics are system-level decisions.

11.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Concurrent calls with the same key share one in-flight request and result.
F2	Calls with different keys run independently.
F3	After completion, the in-flight entry is cleared (next call re-fetches).
F4	A failure is shared by all current waiters (they don't each retry blindly).
F5	Cancellation of one caller must not cancel the shared work for others.
F6	Thread-safe under high concurrency.

Non-functional requirements

Attribute	Policy
Correctness	Exactly one in-flight call per key; shared result.
Performance	O(1) map ops; negligible overhead; big backend-load reduction.
Reliability	Failures don't leave stale in-flight entries (no permanent wedging).
Memory	In-flight map cleared on completion; bounded.
Concurrency	Safe under many coroutines; no lost updates.

Scope

- **In scope:** the single-flight primitive (keyed `Deferred` sharing), failure sharing, cancellation safety, integration with repositories.
- **Out of scope:** result caching/TTL (Ch 12 — single-flight ≠ cache), retry/backoff (Ch 2), token refresh application (Ch 19), debounce (Ch 9).

Assumptions & constraints

- Coroutines; a Hilt `@Singleton` coordinator; keys are stable strings. Single-flight de-dupes *concurrent* calls only; it is **not** a cache (it does not remember results after completion — that's Ch 12).

11.3 System Design Fundamentals

- **Single-flight vs cache.** Single-flight coalesces *concurrent* identical work; a cache remembers results *across time*. They compose: single-flight prevents the stampede when a cache entry expires. Confusing the two is a common interview error.
- **Shared `Deferred`.** The primitive stores an in-flight `Deferred<T>` per key in a map; concurrent callers `await()` the same `Deferred`. One execution, many awaiters.
- **Scope ownership.** The work runs in a **coordinator-owned scope**, not any one caller's scope — so if a single caller cancels, the shared work (and other waiters) survive (F5). The coordinator uses `async` in its own `SupervisorScope`.
- **Keying.** The key identifies "the same request" (e.g., `"user:me"`, `"product:123"`); choosing keys is a design decision (too coarse → wrong sharing; too fine → no dedup).
- **Cleanup.** On completion (success or failure), remove the entry so the next call re-fetches fresh data. Leaving it would turn single-flight into an accidental permanent cache of a stale/failed value.
- **Failure sharing.** All current waiters receive the same failure; they may then retry per policy (Ch 2) — but the single-flight itself doesn't retry.

Core insight: **store one in-flight `Deferred<T>` per key in a mutex-guarded map, run it in the coordinator's own scope, and remove it on completion** — concurrent duplicates collapse to one call while independent keys stay parallel.

11.4 High-Level Architecture



Components & ownership

- **`SingleFlight`** (Hilt `@Singleton`) owns an `inFlight` map keyed by request key, a `Mutex` guarding it, and a coordinator `CoroutineScope` with a `SupervisorJob`.
- **Callers** invoke `singleFlight.run(key) { safeApiCall { ... } }`; the first caller starts the work, the rest await the shared `Deferred`.
- **Networking** (`safeApiCall`, Ch 1) does the actual call once.

Direction: callers → coordinator → one network call → shared result fans out.

11.5 Detailed Component Design



Element	Responsibility	Semantics
`inFlight: Map<Key, Deferred<T>>`	track active work per key	one per key
`Mutex`	guard map read-modify-write	no lost entries
coordinator `scope` (`SupervisorJob`)	run shared work	survives individual caller cancellation
`run(key, block)`	coalesce + await + cleanup	exactly-once per concurrent burst

Caller-cancellation safety: callers `await()` the shared `Deferred`; if a caller's own coroutine is cancelled, its `await()` throws `CancellationException` *for that caller only* — the `Deferred` (owned by the coordinator scope) keeps running for the others. This is the key correctness property.

11.6 End-to-End Data Flow


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11.7 Complete Kotlin Implementation

Package: `com.streamcart.core.network.singleflight`.

11.7.1 The single-flight primitive

```
@Singleton
class SingleFlight @Inject constructor(
    @ApplicationScope private val scope: CoroutineScope,   // SupervisorJob-backed, app-lifetime
) {
    private val inFlight = HashMap<String, Deferred<Any?>>()
    private val mutex = Mutex()

    @Suppress("UNCHECKED_CAST")
    suspend fun <T> run(key: String, block: suspend () -> T): T {
        val deferred: Deferred<T> = mutex.withLock {
            // Existing in-flight work for this key? Share it.
            (inFlight[key] as Deferred<T>?) ?: scope.async {
                try { block() } finally { mutex.withLock { inFlight.remove(key) } }  // cleanup
(F3)
            }.also { inFlight[key] = it as Deferred<Any?> }
        }
        return deferred.await()   // caller-scoped await; cancelling A doesn't cancel the
Deferred
    }
}
```

> **Why `scope.async` (coordinator scope), not `async` in the caller's scope?** If the work ran in the first caller's scope, cancelling that caller would cancel the shared work for everyone (F5 violated). Running in the app-scoped `SupervisorJob` decouples the shared work's lifetime from any single caller.

11.7.2 Providing the application scope (Hilt)

```
@Qualifier annotation class ApplicationScope

@Module @InstallIn(SingletonComponent::class)
object CoroutineScopeModule {
    @Provides @Singleton @ApplicationScope
    fun provideAppScope(@DefaultDispatcher dispatcher: CoroutineDispatcher): CoroutineScope =
        CoroutineScope(SupervisorJob() + dispatcher)
}
```

11.7.3 Repository usage

```
class ProfileRepositoryImpl @Inject constructor(
    private val api: ProfileApi,
    private val single: SingleFlight,
    private val dispatchers: DispatcherProvider,
) : ProfileRepository {

    // Many screens may call this concurrently; only one GET /me goes out.
    override suspend fun me(): Outcome<User> =
        single.run("user:me") {
            safeApiCall(dispatchers.io) { api.me().toDomain() }
        }
}
```

11.7.4 Guarding a non-idempotent mutation against double-tap

```
// Double-tap "Apply coupon" should not send two requests concurrently.
suspend fun applyCoupon(code: String): Outcome<Cart> =
    single.run("apply-coupon:$code") {       // coalesces concurrent identical taps
        safeApiCall(dispatchers.io) { api.applyCoupon(code) }
    }
// NOTE: this only de-dupes CONCURRENT taps. For retries/resumes across time,
// use an idempotency key (Ch 15); single-flight is not a substitute.
```

11.7.5 Module structure

```
:core:network/singleflight  SingleFlight, CoroutineScopeModule, ApplicationScope
:data:profile               ProfileRepositoryImpl (uses SingleFlight)
```

11.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Two callers create entries simultaneously	Two calls for one key	`Mutex` guards check-then-insert (atomic)
Caller A cancels mid-flight	Shared work cancelled for B	Work runs in coordinator scope, not A's
Completion vs new caller race	New caller joins finished work	Entry removed in `finally`; new caller re-fetches
Failure leaves stale entry	Key permanently wedged	`finally { remove }` runs on failure too
Key too coarse	Wrong data shared	Include all distinguishing params in the key
Map grows unbounded	Memory	Entries removed on completion; bounded by concurrent keys

- **The check-then-insert must be atomic** (under one `mutex.withLock`); otherwise two coroutines both see "no entry" and both start work. This is the whole correctness crux.
- **Caller cancellation isolation** is why the work lives in the coordinator scope; test it explicitly (§11.14).

11.9 Failure Scenarios and Recovery

Failure	Detection	Response	Effect on waiters
Network failure	`Outcome.Failure` from block	shared to all waiters; entry removed	all get the same failure; may retry (Ch 2)
Exception in block	`Deferred` completes exceptionally	`await` throws for waiters; entry removed in `finally`	no permanent wedge
Caller cancelled	that `await` throws	only that caller stops	others continue
Coordinator scope cancelled (app shutdown)	all deferreds cancel	app teardown	acceptable
Key collision (too coarse)	wrong result shared	fix key design	correctness bug
Slow call blocks waiters	long await	waiters share the latency (acceptable)	add timeout in block (Ch 1 `callTimeout`)


11.9 Failure Scenarios and Recovery

> **Don't cache failures:** removing the entry in `finally` ensures a failed single-flight doesn't pin a broken result; the next call tries again.

11.10 Scaling the Implementation

- **Level 1.** Use it for a couple of hot reads (`/me` , `/config`).
- **Level 2.** A shared `SingleFlight` singleton used across repositories.
- **Level 3 — production.** Combine with caching (Ch 12): check cache → on miss, single-flight the fetch → populate cache (prevents stampede); add per-call timeouts.
- **Level 4 — large.** Keyed by rich parameters; integrate with token refresh (Ch 19 is a specialized single-flight); metrics on coalesce ratio.
- **Level 5 — enterprise.** A generic request coordinator layer (coalesce + cache + retry + breaker) applied uniformly; per-endpoint coalescing policies.

Avoid overengineering: not every request needs single-flight — apply it to hot, frequently-duplicated reads and to refresh/stampede-prone paths, not to every call.

11.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Dedup scope	Single-flight (concurrent)	Cache (across time)	Use both; they solve different problems
Primitive	Shared `Deferred` in a map	`Mutex` per key (serialize)	Deferred shares one result; per-key mutex serializes but still makes N calls
Scope	Coordinator (SupervisorJob)	First caller's scope	Coordinator so caller cancellation doesn't kill shared work
Result sharing	`async` / `await`	`SharedFlow(replay=1)`	async for request/response; SharedFlow for streams
Mutations	Single-flight (concurrent dedup)	Idempotency key (across time)	Idempotency key for retries/resume; single-flight for double-tap
Library	Hand-rolled	Store/Room's built-ins	Hand-rolled is tiny; libraries (e.g., Store) bundle cache+single-flight

Single-flight vs per-key mutex: a per-key `Mutex` *serializes* duplicate calls — the second waits, then makes its *own* call (so you still do N backend requests, just not in parallel). Single-flight makes the second *share* the first's result — **one** backend request. For dedup you want single-flight.

Single-flight vs cache: single-flight does not remember results; two calls one second apart (non-overlapping) each hit the network. Add a cache (Ch 12) for across-time reuse.

11.12 Security Considerations

- **Key isolation across accounts:** include the account/user id in the key (`"user:me:$accountId"`) so account A's in-flight result never fans out to account B after a fast account switch (Ch 42).
- **Don't key on secrets:** keys may appear in logs/metrics; use ids, not tokens.
- **Shared-failure caution:** a single-flight that shares an auth failure must not mask per-caller authorization differences; keep authz server-side (not keyed client-side).
- Client vs backend: coalescing is a client optimization; the backend still authorizes and rate-limits each logical request independently.

11.13 Performance and Resource Optimization

- **Backend-load reduction:** the headline win — N concurrent duplicates become 1 call; measure the *coalesce ratio* (duplicates saved).
- **Battery/data:** fewer radio activations and parses.
- **Overhead:** a map lookup under a short mutex — negligible; uncontended in practice.
- **Latency:** waiters share the single call's latency (same as a normal fetch); add a per-call timeout (Ch 1 `callTimeout`) so a hung call doesn't block waiters forever.
- **Profiling:** count backend calls vs call sites (Network Inspector) to confirm coalescing; watch the in-flight map size.

11.14 Testing Strategy

Test with controllable suspension to force the concurrent window.

```
@OptIn(ExperimentalCoroutinesApi::class)
class SingleFlightTest {
    private val scope = TestScope()

    @Test fun `concurrent calls for same key execute block once`() = runTest {
        val sf = SingleFlight(backgroundScope)
        val calls = AtomicInteger(0)
        val gate = CompletableDeferred<Unit>()
        val block: suspend () -> String = { calls.incrementAndGet(); gate.await(); "ok" }

        val a = async { sf.run("k", block) }
        val b = async { sf.run("k", block) }
        runCurrent()                    // both enter run("k")
        gate.complete(Unit)
        assertEquals("ok", a.await()); assertEquals("ok", b.await())
        assertEquals(1, calls.get())    // executed ONCE
    }

    @Test fun `different keys run independently`() = runTest {
        val sf = SingleFlight(backgroundScope); val calls = AtomicInteger(0)
        awaitAll(async { sf.run("a") { calls.incrementAndGet(); "x" } },
                 async { sf.run("b") { calls.incrementAndGet(); "y" } })
        assertEquals(2, calls.get())
    }

    @Test fun `cancelling one caller does not cancel shared work`() = runTest {
        val sf = SingleFlight(backgroundScope); val done = CompletableDeferred<String>()
        val a = async { sf.run("k") { done.await() } }
        val b = async { sf.run("k") { done.await() } }
        runCurrent()
        a.cancel()                      // caller A bails
        done.complete("ok")
        assertEquals("ok", b.await())   // B still gets the result
    }

    @Test fun `entry cleared after completion so next call refetches`() = runTest {
        val sf = SingleFlight(backgroundScope); val calls = AtomicInteger(0)
        sf.run("k") { calls.incrementAndGet(); "a" }
        sf.run("k") { calls.incrementAndGet(); "b" }   // sequential, not concurrent
        assertEquals(2, calls.get())    // not cached across time
    }
}
```

Also: failure is shared and entry removed (next call retries); high-concurrency stress (100 coroutines, assert 1 call).

11.15 Production Observability

- **Metrics:** coalesce ratio (requests coalesced / total), in-flight map size, single-flight failure rate, time-to-share.
- **Logs:** `single_flight{key_type, coalesced_count}` (key *type*, not raw key if it can carry PII), `single_flight_error{key_type}`.
- **Debugging:** backend sees N identical concurrent calls → single-flight not applied or key too fine; wrong user's data shown → key missing account id; "stuck loading" → entry not removed on failure (missing `finally`).
- **Privacy:** never log raw keys that embed PII; log key categories.

11.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *What is single-flight?* Coalescing concurrent identical requests into one in-flight call whose result is shared.
2. *Single-flight vs cache?* Single-flight de-dupes *concurrent* work; cache remembers *across time*. Different problems; compose them.
3. *What's a cache stampede and how does single-flight help?* Many simultaneous cache misses for one key → single-flight makes them share one fetch.
4. *Why not just a per-key mutex?* It serializes but still makes N calls; single-flight shares one result.
5. *What primitive shares the result?* A single `Deferred<T>` per key that all callers `await`.

Architecture

6. *Where does the shared work run?* A coordinator-owned `SupervisorJob` scope, not any caller's scope.
7. *Why?* So one caller's cancellation doesn't cancel the shared work for others.
8. *How is the key chosen?* It identifies "the same request" — include all distinguishing params (and account id).
9. *Is it a singleton?* Yes — shared across repositories.

Kotlin/concurrency

10. *Why must check-then-insert be atomic?* Otherwise two coroutines both see "no entry" and both start work (mutex guards it).
11. *How is the entry cleaned up?* `finally { remove(key) }` — on success *and* failure.
12. *What happens to waiters on failure?* They share the failure; entry removed so the next call retries.
13. *Caller cancels — what do others see?* Nothing changes; only the cancelled caller's `await` throws.

Correctness

14. *Does single-flight cache results?* No — non-overlapping calls each fetch.
15. *Can it dedupe a double-tapped mutation?* Concurrent taps yes; retries across time need an idempotency key (Ch 15).
16. *Stale in-flight entry risk?* Prevented by `finally` cleanup.

Integration

17. *How does it relate to token refresh?* Refresh is a specialized single-flight (one refresh for many 401s, Ch 19).
18. *How does it combine with a cache?* Cache check → miss → single-flight fetch → populate cache (anti-stampede).

Security

19. *Cross-account leakage?* Include account id in the key; clear on switch.
20. *Keys and secrets?* Don't key on tokens; keys may be logged.

Performance

21. *Main benefit?* Fewer backend calls/parses/radio activations.
22. *Overhead?* A map op under a short mutex — negligible.
23. *How measure effectiveness?* Coalesce ratio; backend calls vs call sites.

Testing

24. *How test one-execution?* Gate the block; launch concurrent callers; assert one invocation.
25. *How test caller-cancellation isolation?* Cancel one; assert the other still completes.
26. *How test no-cache-across-time?* Two sequential calls → two executions.

Design-change challenges

1. **Cache + single-flight to kill a stampede.** Combine as above (Ch 12).
2. **Per-account correctness.** Key includes account id; clear on switch (Ch 42).
3. **Specialize for token refresh.** One refresh for all 401s; version tokens (Ch 19).
4. **Add a timeout so a hung call doesn't block waiters.** `callTimeout` (Ch 1) / `withTimeout` in the block.
5. **Coalesce across process via WorkManager?** Unique work (Ch 30) for background dedup.

Mock walkthrough (abridged)

Clarify (concurrent dup problem? cache too?) → shared `Deferred` map + mutex → coordinator scope for cancellation safety → cleanup on completion → keys + account isolation → combine with cache → testing concurrency.

11.17 Practical Exercises

1. **Beginner.** Implement `SingleFlight.run(key, block)`; prove 100 concurrent same-key calls execute the block once. *Eval:* 1 execution; shared result.
2. **Intermediate.** Prove caller-cancellation isolation and post-completion cleanup. *Eval:* survivor completes; next call refetches.
3. **Advanced.** Combine with an LRU cache to prevent a stampede on expiry; add a per-call timeout. *Eval:* 1 fetch on mass miss; no hang.
4. **Production debugging.** Fix a repo issuing N identical `/me` calls on launch (no single-flight) and one leaking account A's result to B (key missing id). *Eval:* 1 call; correct per-account.

5. **System design challenge.** Turn token refresh into a single-flight so 50 simultaneous 401s trigger one refresh (Ch 19). *Eval:* one refresh; all retry with the new token.

(Solutions withheld unless requested.)

11.18 Chapter Summary & Reference Sheet

Key decisions: store one in-flight `Deferred<T>` per key in a **mutex-guarded map**; run the work in a **coordinator `SupervisorJob` scope** (so caller cancellation doesn't kill shared work); **remove the entry on completion** (no across-time caching, no stale/failed wedging); **key includes account id**; single-flight **is not a cache** (combine with Ch 12) and **not** an idempotency substitute (Ch 15).

Concepts: request coalescing, shared `Deferred`, cache stampede/thundering herd, coordinator scope, caller-cancellation isolation, key design.

Common mistakes: non-atomic check-then-insert (two calls); running in the caller's scope (cancellation kills shared work); not removing the entry (stale cache / wedge); treating single-flight as a cache or as idempotency; keys missing account id.

Checklist:

- [ ] mutex-guarded map
- [ ] coordinator SupervisorJob scope
- [ ] shared `Deferred.await`
- [ ] `finally` cleanup
- [ ] account id in key
- [ ] per-call timeout
- [ ] combine with cache for stampede
- [ ] no PII in keys/logs.

Key APIs: `CoroutineScope(SupervisorJob())`, `async` / `await`, `Mutex.withLock`, `Deferred`, `CompletableDeferred` (tests), `TestScope` / `backgroundScope`, `runCurrent`.

References:

- Coroutines, async/await: https://kotlinlang.org/docs/composing-suspending-functions.html
- Shared mutable state & Mutex: https://kotlinlang.org/docs/shared-mutable-state-and-concurrency.html
- CoroutineScope / SupervisorJob: https://kotlinlang.org/docs/coroutine-context-and-dispatchers.html
- (concept) Go `singleflight`: https://pkg.go.dev/golang.org/x/sync/singleflight
- Store (cache + single-flight): https://github.com/MobileNativeFoundation/Store

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; full primitive + DI + usage + mutation-guard; concurrency (atomic insert, caller isolation); failure/cleanup; 5 levels; single-flight-vs-cache-vs-mutex trade-offs; security (account keying); perf (coalesce ratio); concurrency tests; observability; 26 Q&A + 5 challenges. ✅

- **Expand later:** cache integration (Ch 12), token refresh (Ch 19), idempotency (Ch 15), Store library.
- **Shared primitive introduced:** `SingleFlight` — reused by Ch 12/19.